circle collapses, causing the fabric of space to pinch. We can repair the pinch by allowing the fabric to momentarily tear, and then replacing the pinched one-dimensional sphere—the collapsed circle—with a zero-dimensional sphere-two points-plugging the holes in the upper and lower portions of the shape arising from the tear. As shown in Figure 13.3, the resulting shape looks like a warped banana, which through gentle deformation (non-space tearing) can be reshaped smoothly into the surface of a beach ball. We see, therefore, that when a one-dimensional sphere collapses and is replaced by a zero-dimensional sphere, the topology of the original doughnut, that is, its fundamental shape, is drastically altered, In the context of the curled-up spatial dimensions, the space-tearing progression of Figure 13.3 would result in the universe depicted in Figure 8.8 evolving into that depicted in Figure 8.7.

Although this is a lower-dimensional analogy, it captures the essential features of what Morrison and I foresaw for the second half of Strominger's story. After the collapse of a three-dimensional sphere inside a Calabi-Yau space, it seemed to us that space could tear and subsequently repair itself by growing a two-dimensional sphere, leading to far more drastic changes in topology than Witten and we had found in our earlier work (discussed in Chapter 11). In this way, one Calabi-Yau shape could, in essence, transform itself into a completely different Calabi-Yau shape-much like the doughnut transforming into the beach ball in Figure 13.3—while string physics remained perfectly well behaved. Although a picture was starting to emerge, we knew that there were significant aspects that we would need to work out before we could establish that our second half of the story did not introduce any singularities—that is, pernicious and physically unacceptable consequences. We each went home that evening with the tentative elation that we were sitting on a major new insight.

A Flurry of E-Mail

The next morning I received an e-mail from Strominger asking me for any comments or reactions to his paper. He mentioned that "it should tie in somehow with your work with Aspinwall and Morrison," because, as it turned out, he too had been exploring a possible connection to the phenomenon of topology change. I immediately sent